

Measuring Go puzzle difficulty

How much attempt data does it take to measure a puzzle's difficulty, instead of assigning it by hand?

What I did. Plant 300 puzzles and 3,000 players with known difficulties and skills. Simulate their first-attempt logs under three exposure patterns; the one modelling Go Magic is skill-tree gating, where a player meets only puzzles near their own level. Fit Glicko-2 with the truth hidden, then score the fitted difficulties against it. Scale: ± 100 points $\approx$ one Go rank (EGF convention, accurate at 1d and generous below); assigning every puzzle the same difficulty scores 467.

About 160 first attempts measure a puzzle's difficulty to roughly one rank, but on skill-tree data only if the log is fitted jointly. Replaying the same log one attempt at a time, the standard online way, leaves almost three ranks of error: 287 points against 112, a 61% cut from how the fit is run alone. The failure mode it avoids: reading 290 points of error off an online refit and concluding the data is inadequate.



simulation: 300 puzzles, 3000 players, 10 reps; planted truth $\sim N(1500, 467) = 20k-9d$ at ± 3 sd. Online Glicko-2 with Lichess clamps. Error is RMSE after removing a mean offset, so compression counts; bands are 95% t intervals. Shows what measuring difficulty would take, not that any label is wrong.

Online Glicko-2 under three exposure patterns: ungated random pairing (a diagnostic test), skill-tree gating as implied by Go Magic's public tree, and gating with 10% of traffic ungated. Dotted line is one-rank accuracy; bands are 95% intervals over ten planted worlds.

Reproduce. `./src/recovery.py --puzzles 300 --reps 10` draws the curves; `./src/batch_fit.py` refits the identical log jointly. Every other number here is one command in `docs/RUNNING.md`; the estimator is checked against Glickman's worked example.

WHAT TO DO ABOUT IT

Backfill jointly, serve online

Fit the history once, jointly, for the labels; keep online Glicko-2 for the live path, where one-at-a-time updating is right.

Ship labels per puzzle, not per catalogue

Coverage will never be uniform: publish a measured label only where the attempts exist. But not on Glicko's own RD, which under gating claims 3.7 $\times$ more precision than it delivers.

Anchor the scale with an ungated test

An ungated diagnostic spanning the rank range is the standard repair for a compressed scale: 25% of traffic through one buys back 57 points and much of the compression.

What this does not claim. Not that any hand-assigned label is wrong; testing that needs the private attempt log. It answers the prior question: if difficulty were measured from attempts, how much data would the answer need to mean anything? That depends only on the estimator and the *shape* of the data, so simulation settles it, and nothing here uses private data.

The next step is small: run the joint fit over one month of the real log, first attempts only, and check the recovery curve against it.

THREE RESULTS

Gating itself costs accuracy

Under gating no attempt directly compares an easy puzzle with a hard one: 1.6 $\times$ the error at 10 first attempts per puzzle, 2.7 $\times$ at 160, and a smaller check holds $\sim 2.8\times$ out to 1,280. Traffic does not close it.

The damage is to the spacing, not the ordering

At 160 attempts the fit still ranks puzzles well (Spearman 0.94), but compresses the scale to under half its width. It knows *which* puzzle is harder, not *by how much*. Compression is repairable.

Uneven traffic is a second, separate cost

Everybody attempts the first node; few reach the last. At matched volume that funnel costs 30–50 points and drops the ordering from 0.94 to 0.78, the one thing gating had left intact.

first attempts per puzzle	ungated	gated, online	gated, joint fit
10	250	408	388
40	167	355	252
160	107	287	112

RMSE in rating points, mean offset removed, since neither estimator fixes an origin ($\pm 100 \approx$ one rank). Ten planted worlds; the last two columns fit the identical log, so only the estimator differs. 95% intervals $\pm 3-8$; every contrast significant.